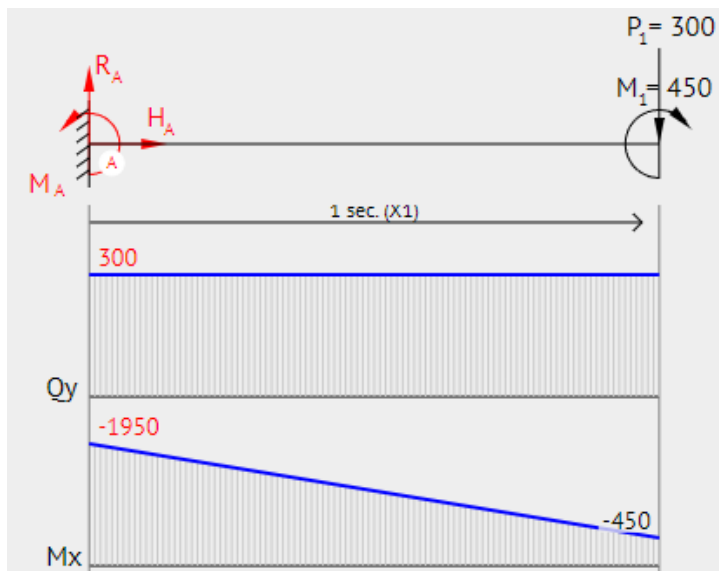
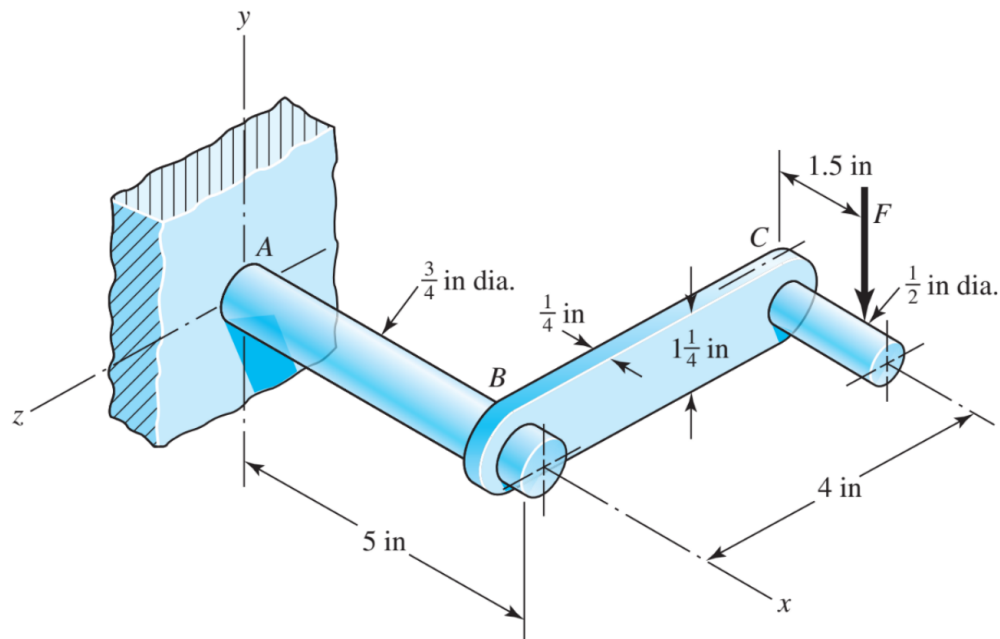


1. A crank loaded by a force $F = 300$ lb causes twisting and bending of a $\frac{3}{4}$ -in-diameter shaft fixed to a support. For point A on the top surface of the shaft: (14' in total)

1) determine the maximum normal and absolute maximum shear stresses. (6')

2) determine the Tresca Stress and the von Mises Stress. (4')

3) use the maximum shear stress theory to determine a specific material type for SAE 3140 steel with safety factor $N = 4$. (4')



$$Torque = 4 * 300 = 1200$$

$$Z_p = \frac{\pi D^3}{16} = \frac{\pi 0.75^3}{16} \cong 0.0828$$

$$\tau = \frac{T}{Z_p} = \frac{1200}{0.0828} \cong 14486.6$$

$$S = \frac{\pi D^3}{32} = \frac{\pi 0.75^3}{32} \cong 0.0414$$

$$\sigma_{max} = \frac{M}{S} = \frac{1950}{0.0414} = 47081.56[psi]$$

$$\sigma_1 = \frac{(\sigma_x + \sigma_y)}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \frac{(47081.56)}{2} + \sqrt{\left(\frac{47081.56}{2}\right)^2 + 14486.6^2} = 51181.87$$

$$\sigma_2 = \frac{(\sigma_x + \sigma_y)}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \frac{(47081.56)}{2} - \sqrt{\left(\frac{47081.56}{2}\right)^2 + 14486.6^2} = -4100.3$$

$$\tau_{abs\ max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{51181.87 - -4100.3}{2} = 27641.085[psi]$$

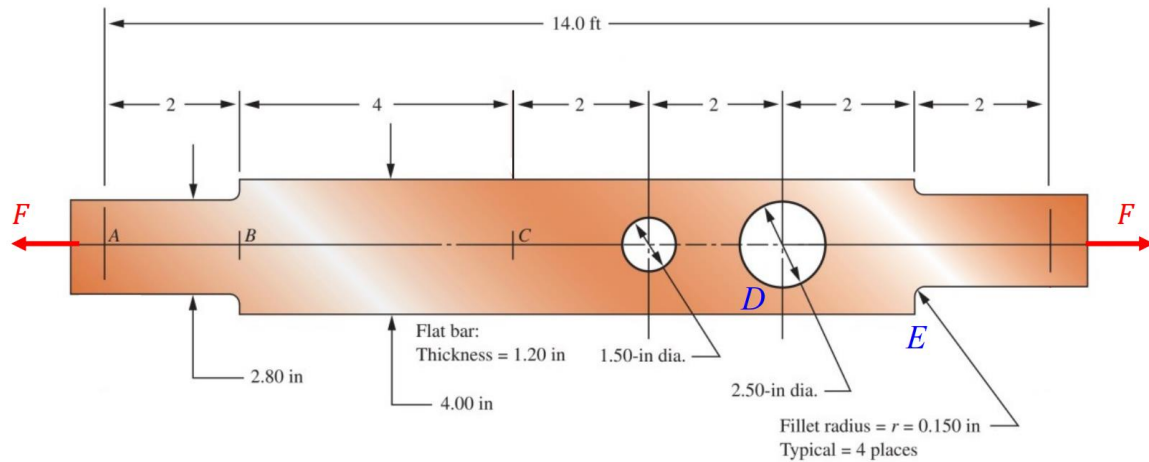
$$\sigma' = \frac{\sigma_1}{2} = \frac{51181.87}{2} = 25590.9375[psi]$$

$$\begin{aligned} \sigma_e &= \sqrt{\frac{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2}{2}} \\ &= \sqrt{\frac{(51181.87 - -4100.3)^2 + (-4100.3)^2 + (-51181.87)^2}{2}} = 53350[psi] \end{aligned}$$

$$\frac{\sigma_1}{2} \leq \frac{S_y}{2N} \rightarrow 4 * 51181.7[psi] \leq S_y$$

MATERIAL = SAE 3140 OQT 400

2. The beam shown is a stepped, flat bar having a constant thickness of 1.20 in. It is subjected to a repeated tensile force in the range from 7200 lb to 14400 lb. Use Goodman criterion to compute the resulting design factor for the right big hole D (2.50-in diameter) and fillet at point E. The beam is made from SAE 1040 Wrought steel (surface condition: polished). Take reliability as 90%. Stress concentration needs to be considered. (14' in total)



$$\sigma_a$$

From Appendix 3, we get that Tensile Strength, $s_u = 72 \text{ ksi}$; $s_y = 42$

From Figure 5-11A we get that Endurance Limit, $s_n \cong 36 \text{ ksi}$

Size Factor: Because we are using Direct/Axial Tension, we get that $C_s = 1$

Reliability Factor: From Table 5-3, we get that $C_R = 0.90$

Type-of-Stress Factor: For axial tension, we get $C_{st} = 0.80$

Material Factor: For wrought Steel, we get $C_m = 1$

$$s'_n = s_n C_m C_{st} C_R C_s = 36 * 1 * 0.8 * 0.9 * 1 = 25.92 \text{ ksi}$$

D - Hole, $K_t = 2.12$

$$\sigma_{max} = \frac{14400}{1.2 * (4 - 0.75)} = 3692.3$$

$$\sigma_{min} = \frac{7200}{1.2 * (4 - 0.75)} = 1846.1$$

$$\sigma_m = 2769.2$$

$$\sigma_a = 923.07$$

$$N = \frac{1}{\left(\frac{K_t \sigma_a}{s'_n} + \frac{\sigma_m}{s_u}\right)} = \frac{1}{\left(\frac{2.12 * 923.07}{25920} + \frac{2769.2}{7200}\right)} = 2.173$$

Fillet, $K_t = 2.6$

$$\sigma_{max} = \frac{14400}{1.2 * (2.8)} = 4285.7$$

$$\sigma_{min} = \frac{7200}{1.2 * (2.8)} = 2142.86$$

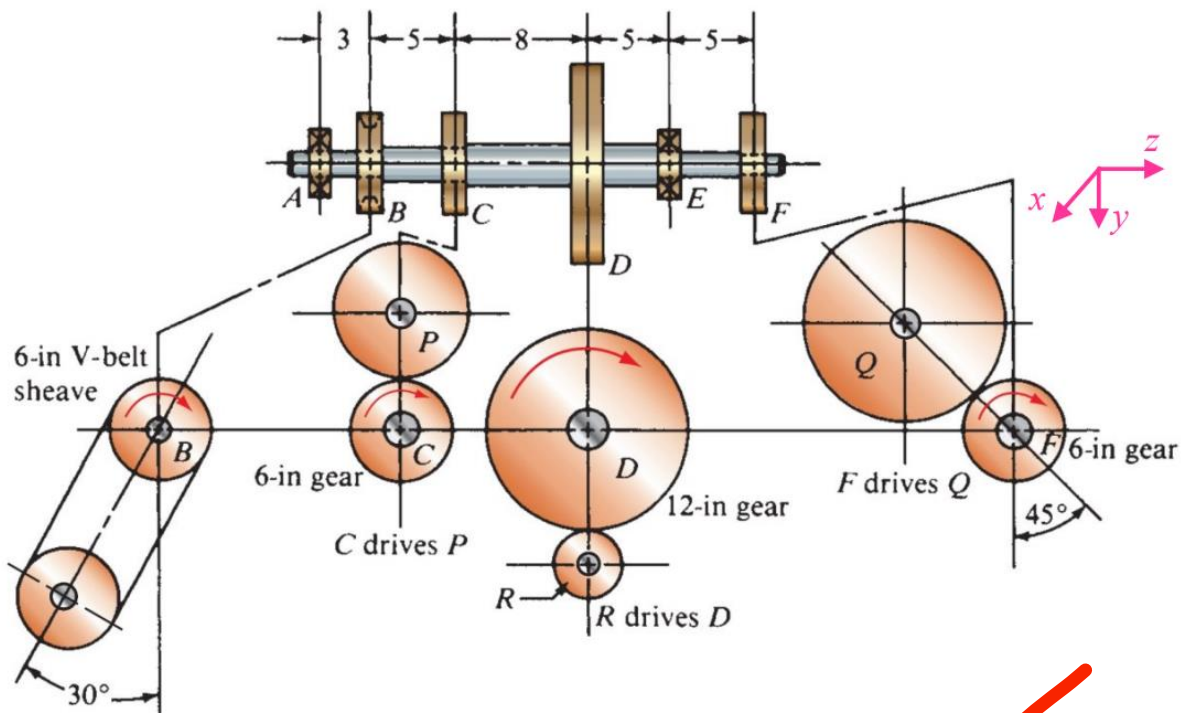
$$\sigma_m = 3214.3$$

$$\sigma_a = 1071.43$$

$$N = \frac{1}{\left(\frac{2.6 * 1071.43}{25920} + \frac{3214.3}{72000}\right)} = 6.574$$

3. All power is input through gear D. Gears C and F drive two separate tool feed devices requiring 5.0 hp each. The V-belt sheave at B requires 2.5 hp to drive a coolant pump. The shaft rotates at 220 rpm. All gears are spur gears with 20°, full-depth teeth. Use SAE 1020 cold-drawn steel for the shaft. Take $C_s = 0.85$, $CR = 0.81$, design factor $N = 5$, stress concentration factor $Kt = 4$. (20' in total)

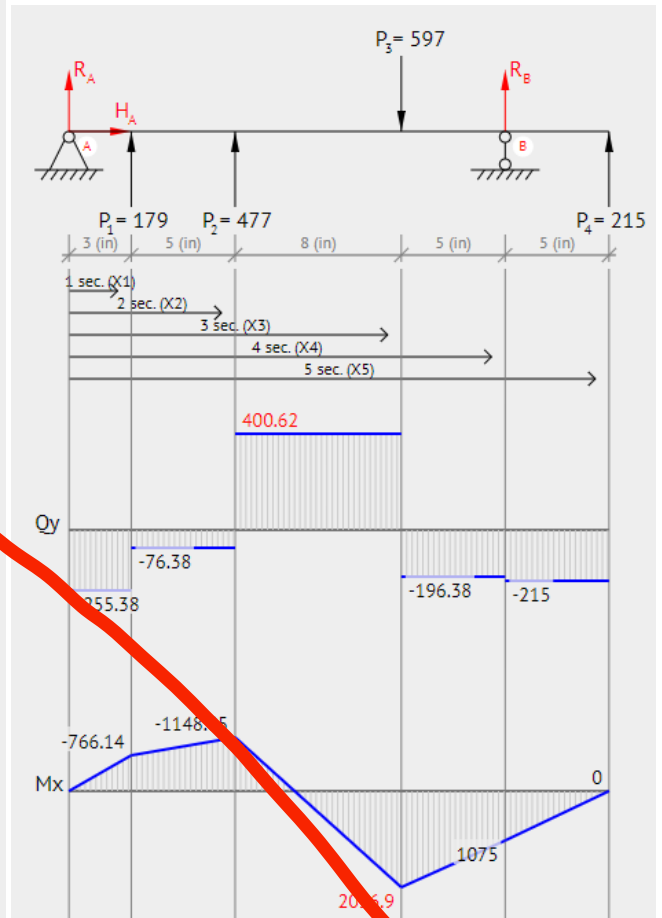
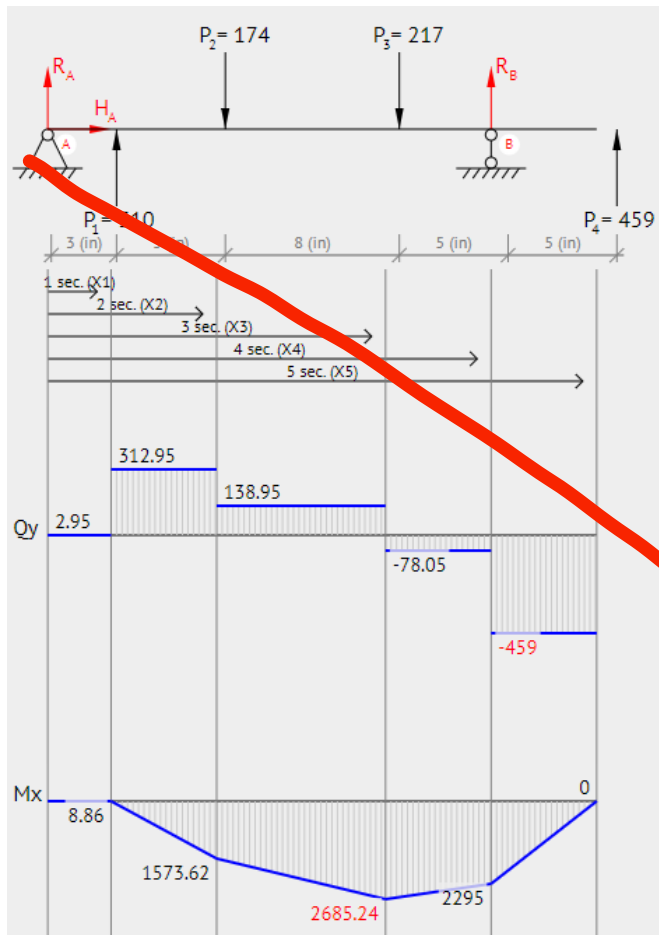
- 1) Determine the torques on V-belt sheave B, gear C, D, and F. Also determine the torque distribution throughout the entire shaft. (3')
- 2) Plot the shear and bending moment diagrams of the entire shaft in the horizontal and vertical planes. (9')
- 3) Determine the total bending moment at point C and D. (2')
- 4) Determine the minimum acceptable diameters for the shaft at point C. Do the design to the left and to the right separately. (6')



$$P_C = 5; P_D = 12.5; P_F = 5; P_B = 2.5$$

$$T_B = 715.9; T_C = 1431.8; T_D = 3579.5; T_F = 1431.8;$$

$$F_{Bx} = 179; F_{By} = 310; W_{Ct} = 477; W_{Cr} = -174; W_{Dt} = -597; W_{Dr} = -217; F_{Ex} = 215; F_{Ey} = 459$$



M(C)=1158

M(D)=2883

DLEFT=2.23

DRIGHT=2.24

-2

4. A spring needs to measure 0 to 1000 N over a deformation scale of length 80 mm. The spring is composed of 30 total coils of a wire with a diameter of 6 mm. The total length of the spring is 200 mm. The wire material is Music wire A228. The ends of the spring are squared and ground. (14' in total)

1) Determine the outer diameter of the spring. (4')

2) Determine the maximum shear stress and the deflection of the spring when a static load of 200 N is applied. (4')

3) Determine the external load required to compress the spring to the solid length; also determine the corresponding maximum shear stress in the spring at the solid length. (6')

Given: $L_f = 200$; $F = 1000$; $f = 80$; $D_w = 6\text{mm}$; $N = 30$; $G = 81.7\text{GPa}$; $\tau = 350\text{MPa}$

Trial:

$$C = 8$$

$$8 = \frac{D_m}{6\text{mm}} = D_m = 48\text{mm}$$

$$K = \frac{4C - 1}{4C - 4} + \frac{0.615}{C} = \frac{4 * 8 - 1}{4 * 8 - 4} + \frac{0.615}{8} = 1.18$$

$$\tau = \frac{8KFC}{\pi D_w^2} = \frac{8 * 1.18 * 1000 * 8}{\pi * 0.006^2} = 667\text{MPa}$$

Type equation here.

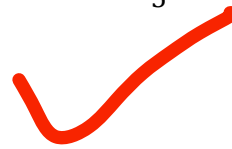
5. Using the boundary-lubricated approach, for a plain surface bearing to carry a radial load of 200 lb from a shaft having a diameter of 0.90 in and rotating at 1800 rpm, determine the minimum possible L/D ratio of the bearing, so that the Manganese bronze (C86200) is acceptable for the bearing material to fulfill the design value of pV factor. (10' in total)

$$\text{design value: } pV = 150000$$

$$pV = 75000$$

$$\frac{F}{LD} * \frac{\pi D n}{12} = 75000 \rightarrow \frac{200}{L * 0.9} * \frac{\pi 0.9 * 1800}{12} = 75000 \rightarrow L = \frac{2\pi}{5} \cong 1.25$$

$$\frac{L}{D} = \frac{\frac{2\pi}{5}}{0.9} = 1.396$$



6. Devise a gear train using all external gears on parallel shafts. Use 20° full-depth involute teeth and no less than 25 teeth and no more than 150 teeth in any gear. The input speed is 6400 rpm and the output speed must be exactly 20 rpm. Ensure that there is no interference. Specify the total number of gear pairs and the number of teeth for each gear pair. (10' in total)

$$TV = \frac{6400}{20} = 320$$

Factors: $2^6, 5$

$$VR_{max} = \frac{150}{25} = 6$$

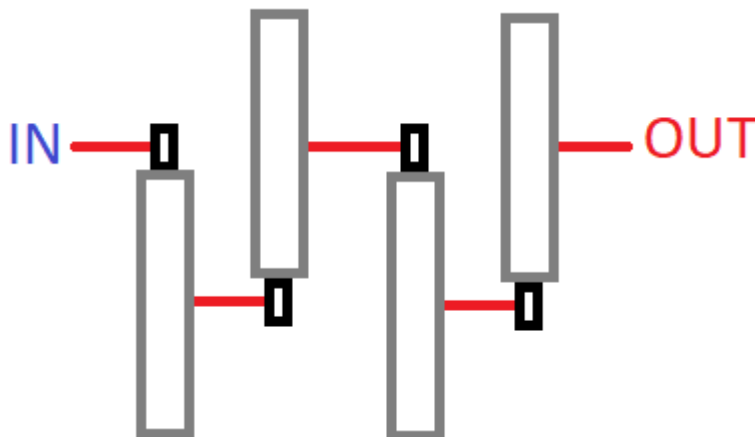
$$6^3 = 216 < 320;$$

$$6^4 = 1296 > 320$$

$N_A=25$	5	$N_A=25$	4	$N_A=25$	4	$N_A=25$	4
$N_B=125$		$N_B=100$		$N_B=100$		$N_B=100$	

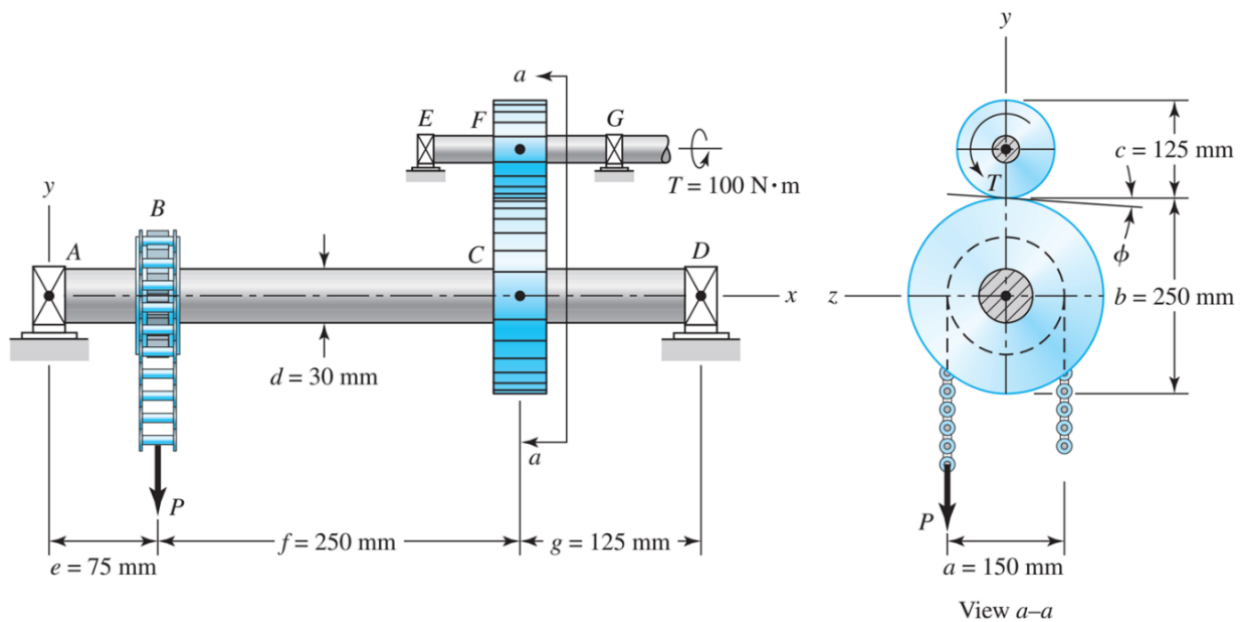


$$\frac{125}{25} * \frac{100}{25} * \frac{100}{25} * \frac{100}{25} = 320$$



7. A torque $T = 100 \text{ N} \cdot \text{m}$ is applied to the shaft EFG. Gear F transmits torque to shaft ABCD through gear C, which drives the chain sprocket at B, transmitting a force P . The sprocket B, gear C, and gear F have pitch diameters of $a = 150$, $b = 250$, and $c = 125 \text{ mm}$, respectively. The contact force between the gears is transmitted through the pressure angle of 20° . The bearings at A, D, E, and G are simple supports. (18' in total)

- 1) Determine the tangential and radial forces that the gear F exerts on the gear C. (4')
- 2) Determine the torque on gear C and sprocket B and determine the force P . (6')
- 3) Determine the bending moment at the point B and C. (8')



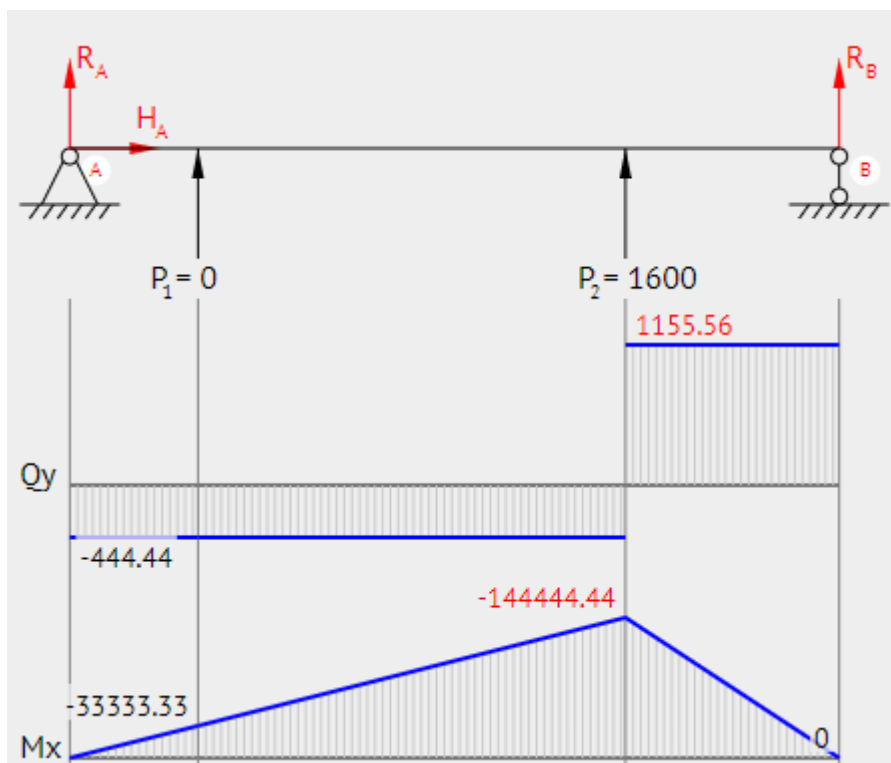
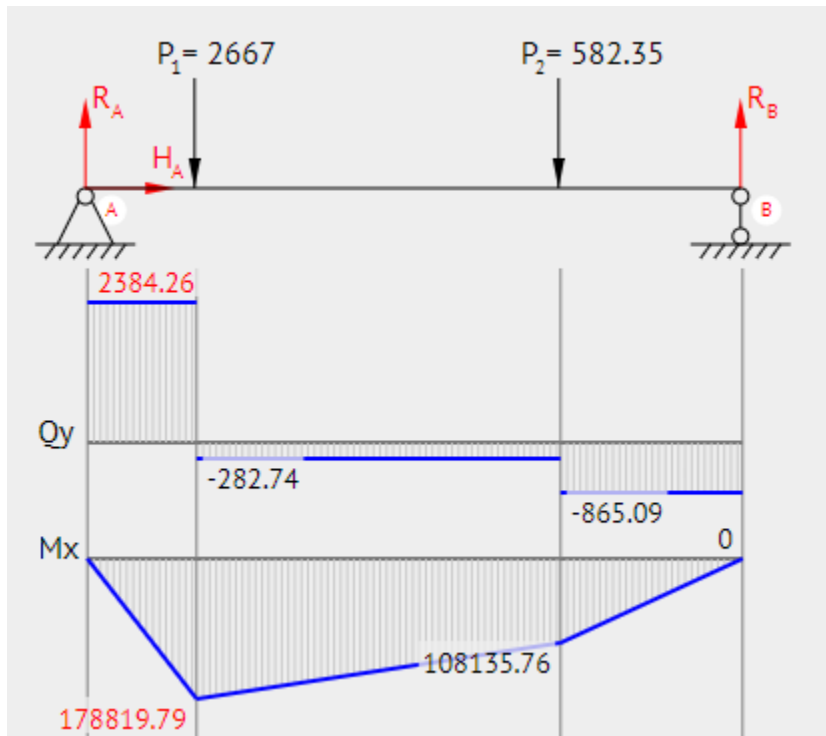
$$F_t \left(\frac{c}{2} \right) = T$$

$$100 = F_t \left(\frac{0.125}{2} \right) \rightarrow F_t = 1600$$

$$F_r = F_t * \tan(\theta) \rightarrow 1600 * \tan(20) = 582.35$$

$$T_B = F_t \left(\frac{b}{2} \right) \rightarrow 1600 \left(\frac{0.25}{2} \right) = 200$$

$$P \left(\frac{a}{2} \right) = T_B \rightarrow \frac{2}{0.15} * 200 = 2666.66$$



$$M_B = \sqrt{333333^2 + 178819.79^2} = 181899.989[N \cdot mm] \cong 181.9[N \cdot m]$$

$$M_C = \sqrt{144444.44^2 + 108135.76^2} = 180437.07[N \cdot mm] \cong 180.4[N \cdot m]$$



Bonus #1: A rectangular steel bar has a cross section 0.40 by 1.20 inch and is 10.5 inch long. The bar has pinned ends and is made of SAE 4140 OQT 700 steel. Compute the allowable load with design factor of 4.

$$E = 11.5E6[psi]$$

$$I = \frac{BH^3}{12} = \frac{0.4 * 1.2^3}{12} = 0.0576$$

$$P_{cr} = \frac{\pi^2 * E * I}{(KL)^2} = \frac{\pi^2 * 11.5 * 10^6 * 0.0576}{(10.5)^2} = 59298.2$$

$$P_A = \frac{P_{cr}}{N} = \frac{59298.2}{4} = 14824.5[lbs]$$

+5